

CS 474: Computer Vision

Underwater Image Enhancement

Implementation, Analysis & Optimization of DIRS-CLAHS

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Assignment 1 • UIEB Dataset • 200 Images • 5 Metrics

AGENDA

Presentation Overview

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- 04 Performance Metrics
- 05 Results & Visual Comparisons
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Problem Background & Dataset

Why underwater images are challenging — and our benchmark



Colour Casts

Red wavelengths absorbed within metres. Blue-green cast dominates raw images.



Low Contrast

Suspended particles scatter light, reducing effective dynamic range.



Blur & Noise

Marine particles and turbulence degrade structural fine detail.

UIEB Dataset

950

Total Images

890

Supervised (with GT)

60

Challenging (no GT)

200

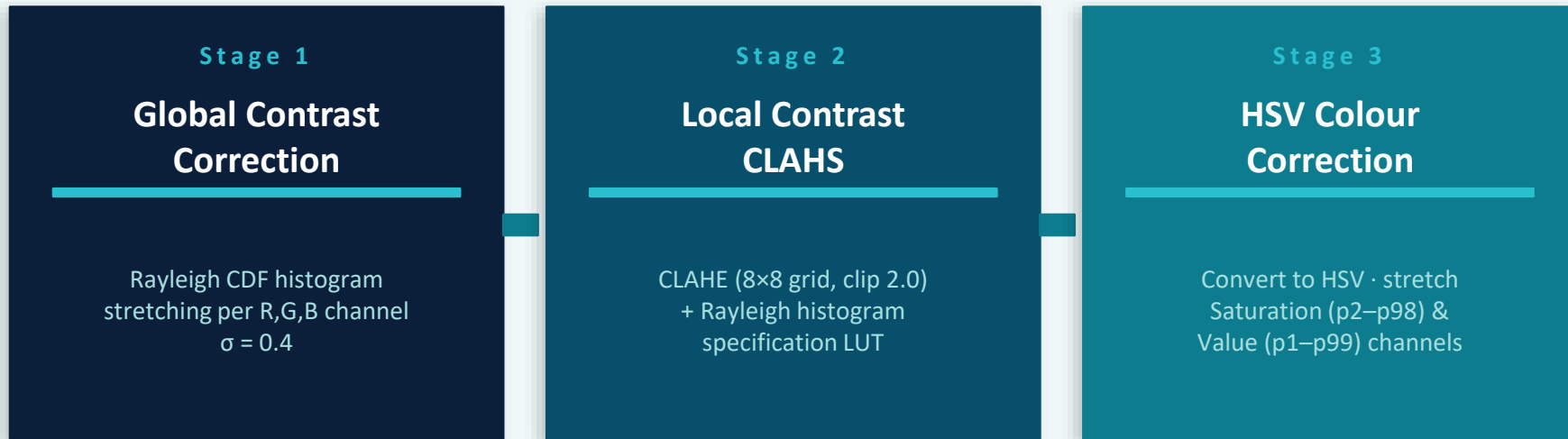
Used in this study

4

Scenario Subsets

DIRS-CLAHS Methodology

Three complementary stages address all major degradation types simultaneously



Key Equations

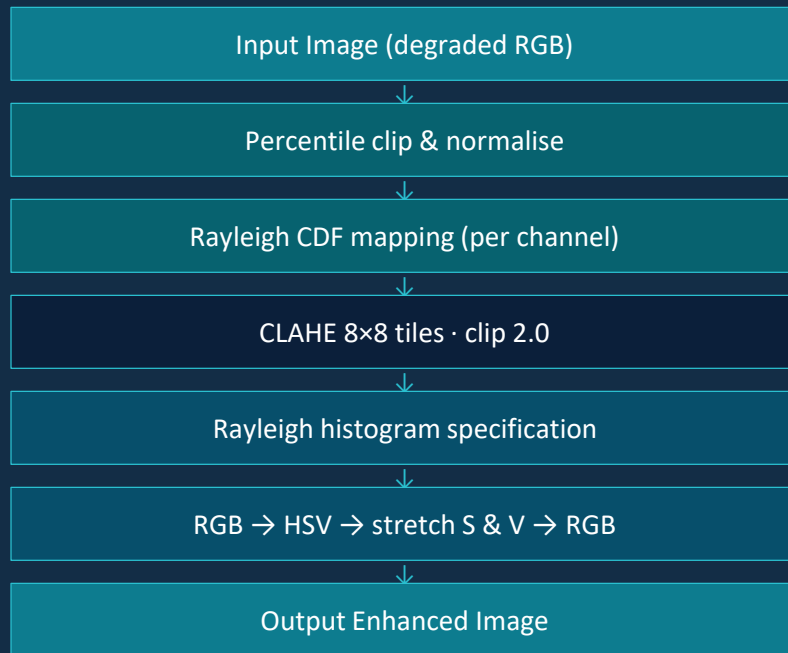
Rayleigh CDF: $F(c^*) = 1 - \exp(-c^{*2} / 2\sigma^2)$ $\sigma=0.4$ **Rayleigh PDF:** $f_R(x) = (x/\sigma^2) \cdot \exp(-x^2/2\sigma^2)$

Implementation Overview

⚙️ Experimental Setup

Platform	Google Colab (CPU only)
Hardware	Intel Xeon @ 2.20 GHz · 12 GB RAM
Language	Python 3.10
Libraries	OpenCV 4.8 · NumPy 1.24 · scikit-image 0.21
Dataset	200 images (seed 42) from 4 UIEB subsets
Params	$\sigma=0.4$ · CLAHE clip=2.0 · grid=8×8

🔄 Processing Pipeline



Evaluation Metrics

FULL-REFERENCE (input_test only)

PSNR

Peak Signal-to-Noise Ratio (dB)

Higher = less distortion. >20 dB acceptable; >30 dB near-lossless.

SSIM

Structural Similarity Index

0–1 scale. Compares luminance, contrast & structure.

MSE

Mean Squared Error

Average squared pixel difference. Lower is better.

NO-REFERENCE (all 4 scenarios)

UCIQE

Underwater Colour Image Quality Evaluation

$0.4680 \cdot \sigma_c + 0.2745 \cdot \text{con}_L + 0.2576 \cdot \mu_s$

Range: 0.3–0.7

UIQM

Underwater Image Quality Measure

$0.0282 \cdot \text{UICM} + 0.2953 \cdot \text{UISM} + 3.5753 \cdot \text{UIConM}$

Range: -2 to +3

Quantitative Results

Full-reference metrics on input_test (N=50) + no-reference across all 200 images

15.307 dB

PSNR (Standard)

↑+0.300 dB with GW-WB

0.6008

SSIM (Standard)

↑+0.0141 with GW-WB

0.4009

UCIQE (avg 200 imgs)

from 0.3339 input

6.89

UIQM (input_test)

from 2.54 input

Per-Scenario No-Reference Metrics

Scenario	UCIQE Input	UCIQE Std	UCIQE +GW	UIQM Input	UIQM Std	UIQM +GW
input_test	0.3191	0.4006	0.4079	2.5392	6.8926	7.6170
input_ce_test	0.3573	0.3833	—	5.8481	8.5487	—
input_gc_test	0.2845	0.4103	—	2.0415	7.1998	—
input_wb_test	0.3747	0.4095	—	5.1960	7.8259	—

Visual Comparisons

Stage-by-stage and cross-scenario qualitative results



Input
(Degraded)



After Rayleigh
Stretch



After CLAHS



After HSV
Correction

Key Visual Observations

- 🔧 Global Rayleigh stretch balances channel histograms — colour cast visibly reduced
- 🔍 CLAHS restores local contrast in both bright foreground and dark background regions
- 🌟 HSV correction boosts saturation, restoring vivid natural underwater colours
- 📊 Enhanced images show good structural fidelity to GT (SSIM $\approx$ 0.60)

Proposed Enhancement: Gray-World White Balance

Stage 0 pre-processing — prepended before Rayleigh stretch

💡 Motivation

In raw underwater images, blue/green channel means are elevated above red due to wavelength absorption. The Rayleigh stretch therefore operates on a biased input, reinforcing colour cast.

📐 Gray-World Formula

$$\mu_{\text{global}} = (\mu_R + \mu_G + \mu_B) / 3$$
$$c_{\text{out}} = \text{clip}(c \times \mu_{\text{global}} / \mu_c, 0, 255)$$

Pipeline Comparison

Standard: Rayleigh Stretch → CLAHS → HSV

Enhanced: GW-WB → Rayleigh → CLAHS → HSV

Improvements (input_test, N=50)

PSNR

+0.300 dB

15.307 → 15.607 dB

SSIM

+0.0141

0.6008 → 0.6149

UCIQE

+0.007

0.4006 → 0.4079

MSE

-346.69

2667.7 → 2321.0

Computational Efficiency

Runtime scalability analysis — CPU benchmark (avg 3 runs)

Resolution	Std (s)	Std FPS	Enh (s)	Enh FPS	Real-Time?
320×240	0.0243	41.2	0.0194	51.4	✓ YES
640×480	0.0696	14.4	0.0628	15.9	✗ No
800×600	0.0943	10.6	0.0965	10.4	✗ No
1280×720	0.1769	5.7	0.2049	4.9	✗ No
1920×1080	0.3994	2.5	0.5610	1.8	✗ No

🚀 Only 320×240 achieves real-time throughput (51.4 fps) on CPU

Runtime scales $\approx O(N^{1.2})$ — slightly superlinear due to patch-based UIConM

- 💡 To achieve 640×480 real-time:
- GPU via CUDA OpenCV (10-20× speedup)
 - Vectorised C++ implementation
 - NumPy interpolation (~40-60% latency)

Key Insights & Discussion

PSNR 15.307 dB is competitive

Traditional methods on UIEB typically achieve 14–19 dB. Deep learning (Water-Net) reaches 22–26 dB — reflecting the fundamental advantage of data-driven training.

UIQM Anomaly: why GW-WB barely changes it

DIRS-CLAHS+GW yields $\Delta \text{UIQM} = 0.0003$ vs standard. GW-WB rescales channel means without altering local spatial structure — so UIConM (patch log-ratio) is unaffected.

Multi-metric evaluation is essential

UCIQE emphasises chroma & saturation (GW-WB helps). UIQM emphasises local sharpness (GW-WB neutral). No single metric captures all perceptual dimensions.

Three-stage architecture is key

Rayleigh alone $\neq$ local fix. CLAHS alone $\neq$ colour cast fix. HSV alone $\neq$ global contrast. Together they address all three degradation axes simultaneously.

Conclusions & Future Work

✓ Conclusions

- DIRS-CLAHS achieves PSNR 15.307 dB, SSIM 0.6008 — competitive with published traditional methods
- UCIQE improves from 0.3339 → 0.4009 across all 200 images (robust generalisation)
- UIQM improves dramatically: input_test 2.54 → 6.89 (standard), 7.62 (GW-WB)
- GW-WB adds +0.300 dB PSNR, +0.0141 SSIM with negligible compute overhead
- Real-time CPU processing at 320×240 (51.4 fps); higher resolutions need GPU

🔭 Future Work

- 1 Vectorised NumPy histogram matching — target 640×480 real-time on CPU
- 2 Evaluate on the challenging 60-image UIEB subset
- 3 Adaptive Rayleigh σ via learning-based regressor (per-image σ selection)
- 4 DIRS-CLAHS as pseudo-label generator to pre-train a lightweight CNN
- 5 Deploy on NVIDIA Jetson Orin for AUV-relevant embedded GPU benchmarks

Thank You!

"DIRS-CLAHS+GW achieves consistent improvement across all metrics with negligible additional computation — a practical drop-in enhancement."